JEE Main Online Exam 2022

Questions & Solution

25th June 2022 | Morning

PHYSICS

Section-A: This section contains 20 multiple choice questions which have only one correct answer. Each question carries **4 marks** for correct answer and **–1 mark** for wrong answer.

Q.1 If $Z = \frac{A^2B^3}{C^4}$, then the relative error in Z will be

(1) $\frac{\Delta A}{A} + \frac{\Delta B}{B} + \frac{\Delta C}{C}$ (2) $\frac{2\Delta A}{A} + \frac{3\Delta B}{B} - \frac{4\Delta C}{C}$ (3) $\frac{2\Delta A}{A} + \frac{3\Delta B}{B} + \frac{4\Delta C}{C}$ (4) $\frac{\Delta A}{A} + \frac{\Delta B}{B} - \frac{\Delta C}{C}$

Ans. **[3]**

Sol. $Z = \frac{A^2B^3}{C^4}$

$\therefore \frac{\Delta Z}{Z} = \frac{2\Delta A}{A} + 3 \times \frac{\Delta B}{B} + \frac{4\Delta C}{C}$

Q.2 $\vec{A}$ is a vector quantity such that $|\vec{A}|$ = non-zero constant. Which of the following expression is true for $\vec{A}$?

(1) $\vec{A} \cdot \vec{A} = 0$ (2) $\vec{A} \times \vec{A} < 0$ (3) $\vec{A} \times \vec{A} = 0$ (4) $\vec{A} \times \vec{A} > 0$

Ans. **[3]**

Sol. $\vec{A} \times \vec{A} = A \times A \times \sin 0°$

$= 0$

Q.3 Which of the following relations is true for two unit vector $\hat{A}$ and $\hat{B}$ making an angle θ to each other?

(1) $|\hat{A} + \hat{B}| = |\hat{A} - \hat{B}| \tan \frac{\theta}{2}$ (2) $|\hat{A} - \hat{B}| = |\hat{A} + \hat{B}| \tan \frac{\theta}{2}$

(3) $|\hat{A} + \hat{B}| = |\hat{A} - \hat{B}| \cos \frac{\theta}{2}$ (4) $|\hat{A} - \hat{B}| = |\hat{A} + \hat{B}| \cos \frac{\theta}{2}$

Ans. **[2]**

Sol. $|\hat{A} - \hat{B}| = 2 \sin\left(\frac{\theta}{2}\right)$

and , $|\hat{A} + \hat{B}| = 2 \cos\left(\frac{\theta}{2}\right)$

$\Rightarrow \frac{|\hat{A} - \hat{B}|}{|\hat{A} + \hat{B}|} = \tan\left(\frac{\theta}{2}\right)$

CAREER POINT

Q.4 If force $\vec{F} = 3\hat{i} + 4\hat{j} - 2\hat{k}$ acts on a particle having position vector $2\hat{i} + \hat{j} + 2\hat{k}$ then, the torque about the origin will be

(1) $3\hat{i} + 4\hat{j} - 2\hat{k}$ (2) $-10\hat{i} + 10\hat{j} + 5\hat{k}$ (3) $10\hat{i} + 5\hat{j} - 10\hat{k}$ (4) $10\hat{i} + \hat{j} - 5\hat{k}$

Ans. **[2]**

Sol. $\vec{\tau} = \vec{r} \times \vec{F}$

$= (2\hat{i} + \hat{j} + 2\hat{k}) \times (3\hat{i} + 4\hat{j} - 2\hat{k})$

$= -10\hat{i} + 10\hat{j} + 5\hat{k}$

Q.5 The height of any point P above the surface of earth is equal to diameter of earth. The value of acceleration due to gravity at point P will be (Given g = acceleration due to gravity at the surface of earth)

(1) $\frac{g}{2}$ (2) $\frac{g}{4}$ (3) $\frac{g}{3}$ (4) $\frac{g}{9}$

Ans. **[4]**

Sol. h = 2R

$\therefore g' = \frac{GM}{(R+h)^2}$

$= \frac{GM}{9R^2}$

$= \frac{g}{9}$

Q.6 The terminal velocity (v_t) of the spherical rain drop depends on the radius (r) of the spherical rain drop as

(1) $r^{1/2}$ (2) r (3) r^2 (4) r^3

Ans. **[3]**

Sol. $6\pi\eta v_t r = \frac{4}{3}\pi r^3 (\rho - \sigma) g$

$\Rightarrow v_t = Cr^2$ where C is a constant

or $v_t \propto r^2$

Q.7 The relation between root mean square speed (v_{rms}) and most probable speed (v_p) for the molar mass M of oxygen gas molecule at the temperature of 300 K will be

(1) $v_{rms} = \sqrt{\frac{2}{3}}\, v_p$ (2) $v_{rms} = \sqrt{\frac{3}{2}}\, v_p$ (3) $v_{rms} = v_p$ (4) $v_{rms} = \sqrt{\frac{1}{3}}\, v_p$

Ans. **[2]**

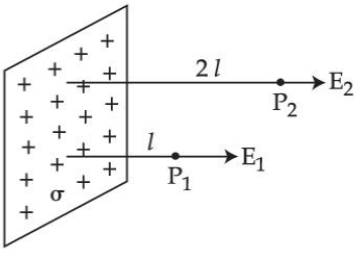
Sol. $v_{rms} = \sqrt{\frac{3RT}{M}}$

$v_p = \sqrt{\frac{2RT}{M}}$

$\Rightarrow v_{rms} = \sqrt{\frac{3}{2}}\, v_p$

$\Rightarrow$ option (2)

Q.8 In the figure, a very large plane sheet of positive charge is shown. P_1 and P_2 are two points at distance ℓ and 2ℓ from the charge distribution. If σ is the surface charge density, then the magnitude of electric fields E_1 and E_2 and P_1 and P_2 respectively are



(1) $E_1 = \frac{\sigma}{\varepsilon_0}$, $E_2 = \frac{\sigma}{2\varepsilon_0}$ (2) $E_1 = \frac{2\sigma}{\varepsilon_0}$, $E_2 = \frac{\sigma}{\varepsilon_0}$ (3) $E_1 = E_2 = \frac{\sigma}{2\varepsilon_0}$ (4) $E_1 = E_2 = \frac{\sigma}{\varepsilon_0}$

Ans. **[3]**

Sol. For an infinite charged plane

$E = \frac{\sigma}{2\varepsilon_0}$ for any value of ℓ

$\Rightarrow E_1 = E_2 = \frac{\sigma}{2\varepsilon_0}$

$\Rightarrow$ option (3)

Q.9 Match List-I with List-II

List-I		List-I	
(1)	AC generator	(I)	Detects the presence of current in the circuit
(2)	Galvanometer	(II)	Converts mechanical energy into electrical energy
(3)	Transformer	(III)	Works on the principle of resonance in AC circuit
(4)	Metal detector	(IV)	Changes an alternating voltage for smaller or greater value

Choose the correct answer from the options given below

(1) (1) - (II), (2) - (I), (3) - (IV), (4) - (III)

(2) (1) - (II), (2) - (I), (3) - (III), (4) - (IV)

(3) (1) - (III), (2) - (IV), (3) - (II), (4) - (I)

(4) (1) - (III), (2) - (I), (3) - (II), (4) - (IV)

Ans. **[1]**

Sol. AC generator $\rightarrow$ Converts mechanical energy into electrical energy

Galvanometer $\rightarrow$ Detects the presence of current in the circuit

Transformer $\rightarrow$ Change AC voltage for smaller or greater value

Metal detector $\rightarrow$ Works on the principle of resonance in AC circuit

$\Rightarrow$ Option (1) is correct

Q.10 A long straight wire with a circular cross-section having radius R, is carrying a steady current I. The current I is uniformly distributed across this cross-section. Then the variation of magnetic field due to current I with distance r (r < R) from its centre will be

(1) $B \propto r^2$ (2) $B \propto r$ (3) $B \propto \frac{1}{r^2}$ (4) $B \propto \frac{1}{r}$

Ans. **[2]**

Sol. $\int \overline{B} \cdot \overline{d\ell} = \mu_0 I_{in}$

$\Rightarrow B \times 2\pi r = \frac{\mu_0 I}{\pi R^2} \pi r^2$

$\Rightarrow B \propto r$

$\Rightarrow$ option (2) is correct

Q.11 If wattles current flows in the AC circuit, then the circuit is :

(1) Purely Resistive circuit (2) Purely inductive circuit

(3) LCR series circuit (4) RC series circuit only

Ans. **[2]**

Sol. For wattless current to flow in AC circuit the circuit will be Purely Inductive circuit

Q.12 The electric field in an electromagnetic wave is given by $E = 56.5 \sin\omega(t - x/c)\ NC^{-1}$. Find the intensity of the wave if it is propagating along x-axis in the free space.

(Given $\in_0 = 8.85 \times 10^{-12} C^2 N^{-1} m^{-2}$)

(1) 5.65 Wm^{-2} (2) 4.24 Wm^{-2} (3) 1.9×10^{-7} Wm^{-2} (4) 56.5 Wm^{-2}

Ans. **[2]**

Sol. $I = \frac{1}{2} \varepsilon_0 E_0^2 c$

$= \frac{1}{2} 8.5 \times 10^{-12} \times (56.5)^2 \times 3 \times 10^8$

$= 4.24\ W/m^2$

Q.13 The two light beams having intensities I and 9I interfere to produce a fringe pattern on a screen. The phase difference between the beams is $\pi/2$ at point P and π at point Q. Then the difference between the resultant intensities at P and Q will be:

(1) 2I (2) 6I (3) 5I (4) 7I

Ans. **[2]**

Sol. $I_p = I + 9I + 2\sqrt{I \times 9I} \cos\frac{\pi}{2} = 10I$

$I_Q = I + 9I + 2\sqrt{I + 9I} \cos\pi = 4I$

So, $I_P - I_Q = 6I$

Q.14 A light wave travelling linearly in a medium of dielectric constant 4, incidents on the horizontal interface separating medium with air. The angle of incidence for which the total intensity of incident wave will be reflected back into the same medium will be :

Given : relative permeability of medium $\mu_r = 1$)

(1) 10° (2) 20° (3) 30° (4) 60°

Ans. **[4]**

Sol. $n = \sqrt{K\mu} = 2$ ($n \Rightarrow$ refractive index)

So for TIR

$\theta > \sin^{-1}\left(\frac{1}{n}\right)$

$\theta > 30$

Only option is 60°

Q.15 Given below are two statements :

Statement I: Davisson-Germer experiment establishes the wave nature of electrons.

Statement II: If electrons have wave nature, they can interfere and show diffraction.

In the light of the above statements choose the correct answer from the option given below :

(1) Both statement I and statement II are true.

(2) Both statement I and statement II are false.

(3) Statement I is true but statement II is false.

(4) Statement I is false but statement II is true.

Ans. **[1]**

Sol. Davisson-Germer experiment is done and establishes the wave nature of electrons.

Interference and diffraction establishes wave nature.

Q.16 The ratio for the speed of the electron in the 3rd orbit of He^+ to the speed of the electron in the 3rd orbit of hydrogen atom will be :

(1) 1 : 1 (2) 1 : 2 (3) 4 : 1 (4) 2 : 1

Ans. **[4]**

Sol. We know that $v \propto \frac{Z}{n}$

$\Rightarrow$ Required ratio = $\frac{\frac{2}{3}}{\frac{1}{3}}$

= 2 : 1

Q.17 The photodiode is used to detect the optical signals. These diodes are preferably operated in reverse biased mode because :

(1) fractional change in majority carriers produce higher forward bias current

(2) fractional change in majority carriers produce higher reverse bias current

(3) fractional change in minority carriers produce higher forward bias current

(4) fractional change in minority carriers produce higher reverse bias current

Ans. **[4]**

Sol. A photodiode is reverse biased. When light falling on it produces charge carriers, the fractional change, in minority carriers is high since the original current is very small.

Q.18 A signal of 100 THz frequency can be transmitted with maximum efficiency by :

(1) Coaxial cable

(2) Optical fibre

(3) Twisted pair of copper wires

(4) Water

Ans. **[2]**

Sol. Optical fibres supports frequency of electromagnetic waves in the range 10^{14} Hz to 10^{15} Hz.

Q.19 The difference of speed of light in the two media A and B, $(\upsilon_A - \upsilon_B)$ is 2.6×10^7 m/s. If the refractive index of medium B is 1.47, then the ratio of refractive index of medium B to medium A is : (Given: speed of light in vacuum C = 3×10^8 ms^{-1})

(1) 1.303 (2) 1.318 (3) 1.13 (4) 0.12

Ans. **[3]**

Sol. Speed of light in a medium = $\frac{c}{n}$

$\Rightarrow$ According to given information

$$\frac{c}{n_A} - \frac{c}{n_B} = 2.6 \times 10^7$$

$$\Rightarrow \frac{n_B}{n_A} - 1 = \frac{2.6 \times 10^7}{3 \times 10^8} \times n_B$$

$$\Rightarrow \frac{n_B}{n_A} \simeq 1.13$$

Q.20 A teacher in his physics laboratory allotted an experiment to determine the resistance (G) of a galvanometer. Students took the observations for $\frac{1}{3}$ deflection in the galvanometer. Which of the below is true for measuring value of G ?

(1) $\frac{1}{3}$ deflection method cannot be used for determining the resistance of the galvanometer.

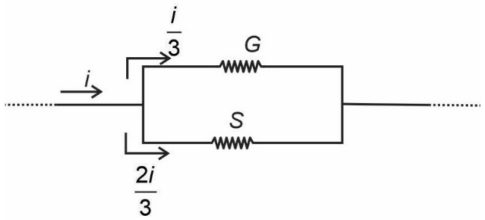
(2) $\frac{1}{3}$ deflection method can be used and in this case the G equals to twice the value of shunt resistance (s)

(3) $\frac{1}{3}$ deflection method can be used and in this case, the G equals to three times the value of shunt resistance (s)

(4) $\frac{1}{3}$ deflection method can be used and in this case the G value equals to the shunt resistance(s)

Ans. **[2]**

Sol. The circuit for the given situation is :



Since G and S are in parallel,

$$\Rightarrow \frac{i}{3} \times G = \frac{2i}{3} \times S$$

$\Rightarrow G = 2S$

$\Rightarrow$ G equals twice the value of shunt resistance.

Section-B: This section contains 10 questions. In Section-B, attempt any **five questions out of 10.** The answer to each of the questions is a numerical value. For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the second decimal place; e.g. 06.25, 07.00, –00.33, –00.30, 30.27, –27.30). Each question carries **4 marks** for correct answer and **–1 mark** for wrong answer.

Q.21 A uniform chain of 6 m length is placed on a table such that a part of its length is hanging over the edge of the table. The system is at rest. The co-efficient of static friction between the chain and the surface of the table is 0.5, the maximum length of the chain hanging from the table is ____m.

Ans. **[2.00]**

Sol. (x) $g\lambda = \mu(6 - x)\, g\lambda$ where x is length of hanging part

$\Rightarrow x = 3 - 0.5\, x$

$\Rightarrow x = 2$ m

Q.22 A 0.5 kg block moving at a speed of 12 ms^{-1} compresses a spring through a distance 30 cm when its speed is halved. The spring constant of the spring will be ___ Nm^{-1}.

Ans. **[600.00]**

Sol. $\frac{1}{2}mV^2 = \frac{1}{2}kx^2 + \frac{1}{2}m\left(\frac{v}{2}\right)^2$

$\Rightarrow \frac{3}{2}mv^2 = \frac{1}{2}kx^2$

$\Rightarrow k = \frac{3}{4} \times \frac{1}{2} \times \frac{144}{9} \times 100$

$\Rightarrow 600$

Q.23 The velocity of upper layer of water in a river is 36 kmh^{-1}. Shearing stress between horizontal layers of water is 10^{-3} Nm^{-2}. Depth of the river is ____ m. (Co-efficient of viscosity of water is 10^{-2} Pa.s)

Ans. **[100.00]**

Sol. $F = -\eta A \frac{du}{dx}$

$\Rightarrow 10^{-3} = 10^{-2} \times \frac{10}{h}$

$\Rightarrow h = \frac{10^{-1}}{10^{-3}}\ m = 100$ m

$\Rightarrow (100)$

Q.24 A steam engine intakes 50 g of steam at 100°C per minute and cools it down to 20°C. If latent heat of vaporization of steam is 540 cal g^{-1}, then the heat rejected by the steam engine per minute is ___ $\times 10^3$ cal. (Given : specific heat capacity of water : 1 cal g^{-1} $°C^{-1}$)

Ans. **[31]**

Sol. $\Delta Q_{rej} = 50 \times 540 + 50 \times 1 \times (100 - 20)$

$= 50 \times [540 + 80]$

$= 50 \times 620$

$= 31000$ cal

$= 31 \times 10^3$ cal

Q.25 The first overtone frequency of an open organ pipe is equal to the fundamental frequency of a closed organ pipe. If the length of the closed organ pipe is 20 cm. The length of the open organ pipe is ___ cm.

Ans. **[80.00]**

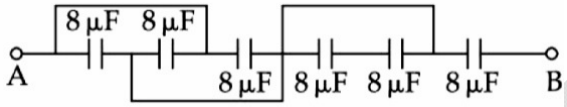
Sol. $2 \times \left(\frac{V}{2L_0}\right) = \left(\frac{V}{4L_C}\right)$

$\Rightarrow L_0 = 4L_C$

$= 4 \times 20$

$= 80$ cm

Q.26 The equivalent capacitance between points *A* and *B* in below shown figure will be _____ μF.



Ans. **[6.00]**

Sol. $C_{eq} = \frac{(3\times8)\times8}{(3\times8)+8}$

$= \frac{24\times8}{32}$

$= 6\ \mu F$

Q.27 A resistor develops 300 J of thermal energy in 15 s, when a current of 2 A is passed through it. If the current increases to 3 A, the energy developed in 10 s is _____ J.

Ans. **[450.00]**

Sol. $300 = I^2R \times 15$

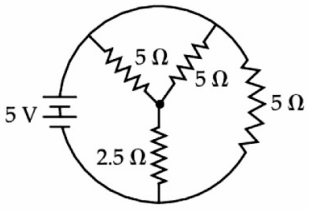
$\Rightarrow R = 5\Omega$

Now $I_2^2Rt_2$

$= 9 \times 5 \times 10$

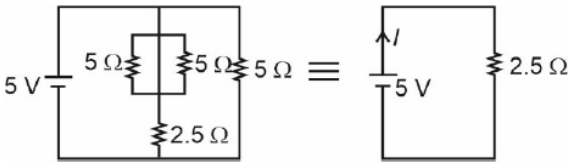
$= 420$ J

Q.28 The total current supplied to the circuit as shown in figure by the 5 V battery is _____A.



Ans. **[2.00]**

Sol. The equivalent circuit is



$\Rightarrow I = \frac{5}{2.5} = 2A$

Q.29 The current in a coil of self inductance 2.0 H is increasing according to $I = 2\sin(t^2)$ A. The amount of energy spent during the period when current changes from 0 to 2 A is _____ J.

Ans. **[4.00]**

Sol. $U = \frac{1}{2}Li^2$

$= \frac{1}{2} 2 \times 2^2 = 4J$

Q.30 A force on an object of mass 100 g is $(10\hat{i} + 5\hat{j})$ N. The position of that object at $t = 2$ s is $(a\hat{i} + b\hat{j})$ m after starting from rest. The value of $\frac{a}{b}$ will be _____.

Ans. **[2.00]**

Sol. $\vec{F} = m\vec{a}$

$\Rightarrow \vec{a} = 100\hat{i} + 50\hat{j}$

So $\vec{S} = \frac{1}{2}\vec{a}t^2$

$= \frac{1}{2}(100\hat{i} + 50\hat{j})2^2$

$= 200\hat{i} + 100\hat{j}$ m

so a = 200 m and b = 100 m

so $\frac{a}{b} = 2$

CHEMISTRY

Section-A: This section contains 20 multiple choice questions which have only one correct answer. Each question carries **4 marks** for correct answer and **–1 mark** for wrong answer.

Q.31 Bonding in which of the following diatomic molecule(s) become(s) stronger, on the basis of MO Theory, by removal of an electron?

(A) NO (B) N_2 (C) O_2 (D) C_2
(E) B_2

Choose the **most appropriate** answer from the options given below :

(1) A, B, C only (2) B, C, E only (3) A, C only (4) D only

Ans. **[3]**

Sol. If an electron is removed from the anti-bonding orbital, then it will tend to increase the bond order.
The HOMO in NO and O_2 is antibonding molecular orbital.
Hence, in NO and O_2 bond order will increase on loss of electron.

Q.32 **Incorrect** statement for Tyndall effect is :

(1) The refractive indices of the dispersed phase and the dispersion medium differ greatly in magnitude.
(2) The diameter of the dispersed particles is much smaller than the wavelength of the light used.
(3) During projection of movies in the cinemas hall, Tyndall effect is noticed.
(4) It is used to distinguish a true solution from a colloidal solution.

Ans. **[2]**

Sol. For Tyndall effect, the diameter of the dispersed particles is not much smaller than the wavelength of the light used.

Q.33 The pair, in which ions are isoelectronic with Al^{3+} is-

(1) Br^- and Be^{2+} (2) Cl^- and Li^+ (3) S^{2-} and K^+ (4) O^{2-} and Mg^{2+}

Ans. **[4]**

Sol. O^{2-}, Mg^{2+} and Al^{3+} are isoelectronic. All have 10 electrons.

Q.34 Leaching of gold with dilute aqueous solution of NaCN in presence of oxygen gives complex [A], which on reaction with zinc forms the elemental gold and another complex [B]. [A] and [B], respectively are :

(1) $[Au(CN_4]^-$ and $[Zn(CN)_2(OH_2]^{2-}$ (2) $[Au(CN_2]^-$ and $[Zn\ (OH_4]^{2-}$

(3) $[Au(CN_2]^-$ and $[Zn(CN)_4]^{2-}$ (4) $[Au(CN_4]^{2-}$ and $[Zn(CN)_6]^{4-}$

Ans. **[3]**

Sol. In the metallurgy of gold

$4Au + 8CN^- + 2H_2O + O_2 \rightarrow 4[Au(CN)_2]^- + 4OH^-$

$2[Au(CN)_2]^- + Zn \rightarrow [Zn(CN)_4]^{2-} + 2Au$

Q.35 Number of electron deficient molecules among the following : PH_3, B_2H_6, CCl_4, NH_3, LiH and BCl_3 is -

(1) 0 (2) 1 (3) 2 (4) 3

Ans. **[3]**

Sol. Only B_2H_6 and BCl_3 are e^- deficient among the given molecules.

Q.36 Which one of the following alkaline earth metal ions has the highest ionic mobility in its aqueous solution?

(1) Be^{2+} (2) Mg^{2+} (3) Ca^{2+} (4) Sr^{2+}

Ans. **[4]**

Sol. In aqueous solution, the ionic mobility is inversely proportional to the charge density on the ion. Hence Sr^{2+} has highest ionic mobility in water.

Q.37 White precipitate of AgCI dissolves in aqueous ammonia solution due to formation of:

(1) $[Ag(NH_3)_4]Cl_2$ (2) $[Ag(Cl)_2(NH_3)_2]$ (3) $[Ag(NH_3)_2]Cl$ (4) $[Ag(NH_3)Cl]Cl$

Ans. **[3]**

Sol. $AgCl + 2NH_3 \rightarrow [Ag(NH_3)_2]Cl$

Q.38 Cerium (IV) has a noble gas configuration. Which of the following is correct statement about it?

(1) It will not prefer to undergo redox reactions.

(2) It will prefer to gain electron and act as an oxidizing agent

(3) It will prefer to give away an electron and behave as reducing agent

(4) It acts as both, oxidizing and reducing agent.

Ans. **[2]**

Sol. E° value of $Ce^{+4}|Ce^{+3}$ is 1.74 V, which suggests that it is a very good oxidising agent.

Q.39 Among the following which is the strongest oxidizing agent?

(1) Mn^{3+} (2) Fe^{3+} (3) Ti^{3+} (4) Cr^{3+}

Ans. **[1]**

Sol. $E^o_{Fe^{3+}|Fe^{2+}} = +0.77$ V $E^o_{Ti^{3+}|Ti^{2+}} = -0.37$ V

$E^o_{Mn^{+3}|Mn^{+2}} = +1.57$ V $E^o_{Cr^{3+}|Cr^{2+}+2} = -0.41$ V

Mn^{+3} is the best oxidising agent among the given series.

Q.40 The eutrophication of water body results in:

(1) loss of Biodiversity.
(2) breakdown of organic matter.
(3) increase in biodiversity.
(4) decrease in BOD.

Ans. **[1]**

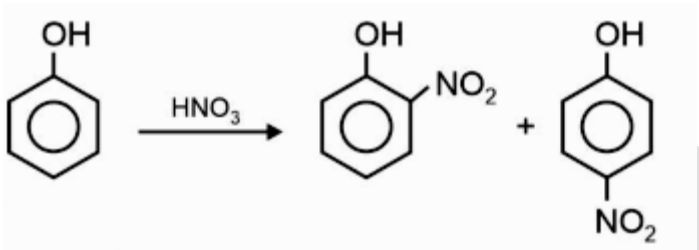
Sol. Eutrophication is the process in which nutrient enriched water bodies support a dense plant population, which kills animal life by depriving it of oxygen and results in subsequent loss of biodiversity.

Q.41 Phenol on reaction with dilute nitric acid, gives two products. Which method will be most efficient for large scale separation?

(1) Chromatographic separation
(2) Fractional crystallisation
(3) Steam distillation
(4) Sublimation

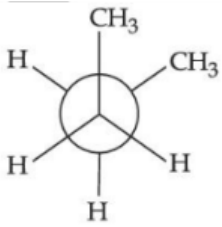
Ans. **[3]**

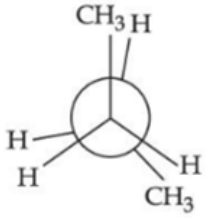
Sol.

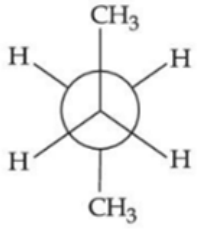


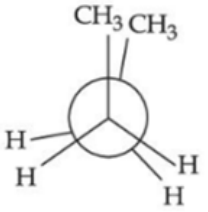
o-Nitrophenol and p-Nitrophenol can be easily separated by steam distillation.

Q.42 In the following structures, which one is having staggered conformation with maximum dihedral angle?

(1)


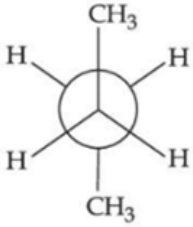
(2)


(3)


(4)


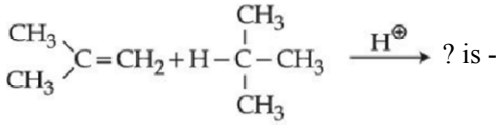
Ans. **[3]**

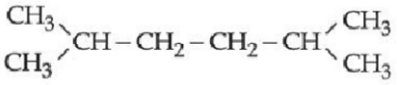
Sol.

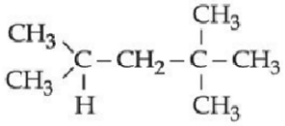


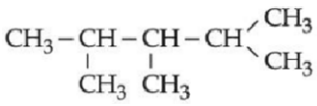
It is the staggered conformation with maximum dihedral angle.

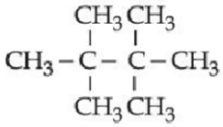
Q.43 The product formed in the following reaction



(1) 

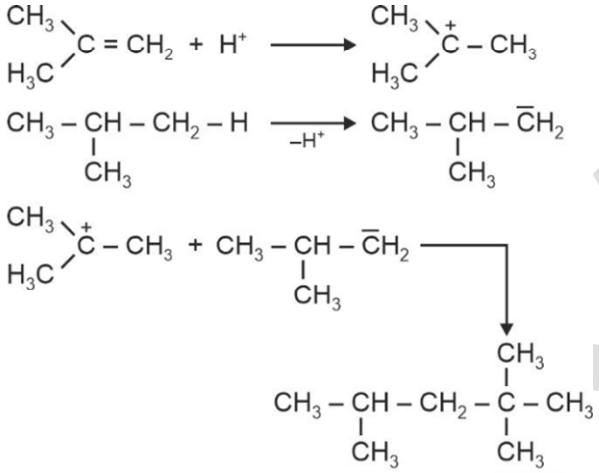
(2) 

(3) 

(4) 

Ans. **[2]**

Sol.



Q.44 The IUPAC name of ethylidene chloride is -

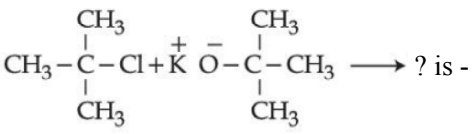
(1) 1-Chloroethene (2) 1-Chloroethyne

(3) 1, 2-Dichloroethane (4) 1, 1-Dichloroethane

Ans. **[4]**

Sol. Ethylidene chloride is $CH_3 - CHCl_2$, its IUPAC name is 1,1-Dichloromethane.

Q.45 The major product in the reaction

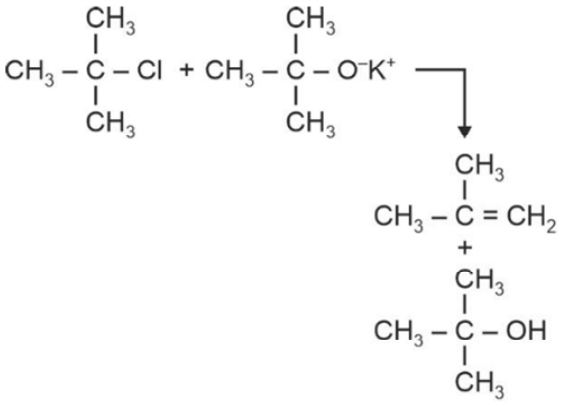


(1) t-Butyl ethyl ether (2) 2, 2-Dimethyl butane

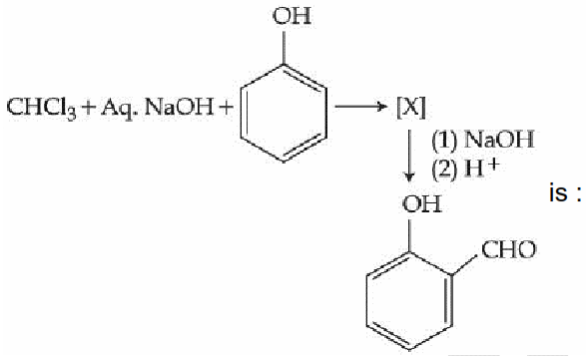
(3) 2-Methyl pent-1-ene (4) 2-Methyl prop-1-ene

Ans. **[4]**

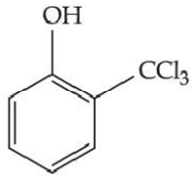
Sol.

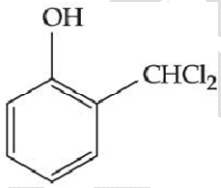


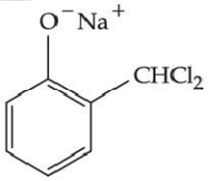
Q.46 The intermediate X, in the reaction :

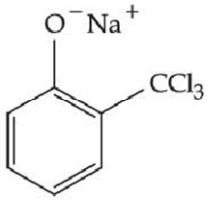


is :

(1) 

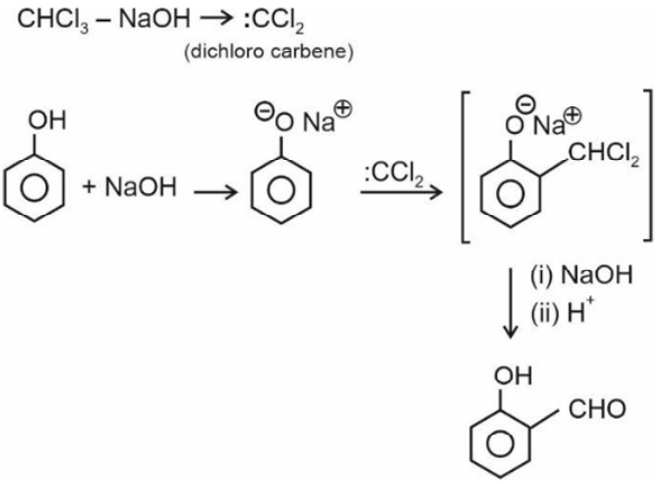
(2) 

(3) 

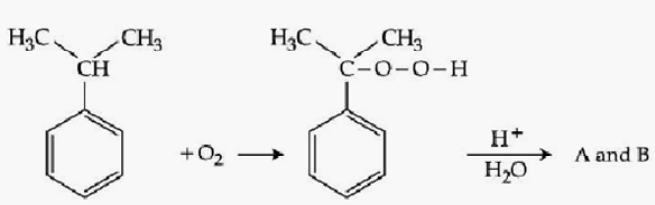
(4) 

Ans. **[3]**

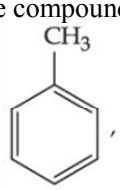
Sol.

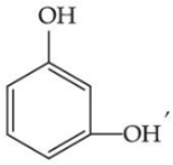


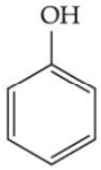
Q.47 In the following reaction:

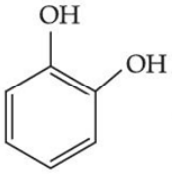


The compound A and B respectively are:

(1)  , CH_3COOH

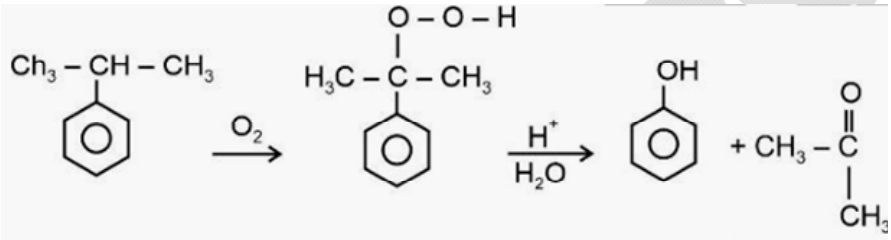
(2)  , CH_3COCH_3

(3)  , CH_3COCH_3

(4)  , CH_3COCH_3

Ans. **[3]**

Sol.

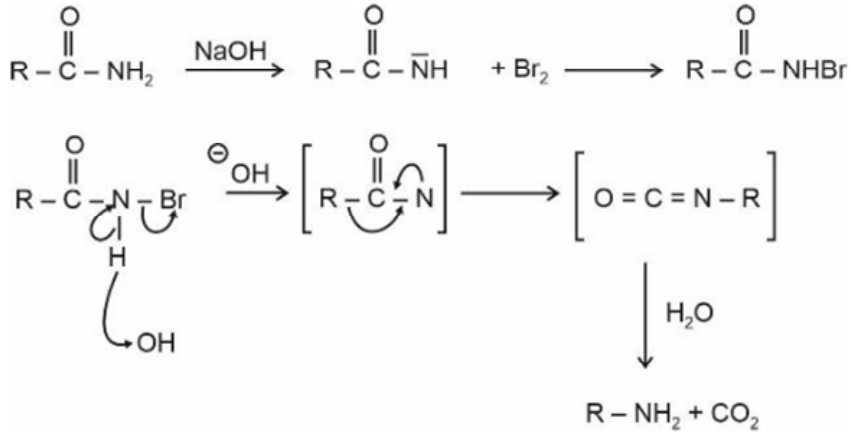


Q.48 The reaction of $R-\underset{\underset{O}{\|}}{C}-NH_2$ with bromine and KOH gives RNH_2 as the end product. Which one of the following is the intermediate product formed in this reaction?

(1) $R-\underset{\underset{O}{\|}}{C}-NH-Br$ (2) $R-NH-Br$ (3) $R-N=C=O$ (4) $R-\underset{\underset{O}{\|}}{C}-NBr_2$

Ans. **[3]**

Sol.



Q.49 Using very little soap while washing clothes, does not serve the purpose of cleaning of clothes, because:
(1) soap particles remain floating in water as ions.
(2) the hydrophobic part of soap is not able to take away grease.
(3) the micelles are not formed due to concentration of soap, below its CMC value.
(4) colloidal structure of soap in water is completely distributed.

Ans. **[3]**

Sol. The little amount of using soap while washing clothes, does not serve the purpose of cleaning of clothes because the micelles are not formed due to the low concentration of soap as it is below CMC.

Q.50 Which one of the following is an example of artificial sweetner ?
(1) Bithional (2) Alitame (3) Salvarsan (4) Lactose

Ans. **[2]**

Sol. Alitame is an artificial sweetner.

Section-B: This section contains 10 questions. In Section-B, attempt any **five questions out of 10.** The answer to each of the questions is a numerical value. Each question carries **4 marks** for correct answer and **–1 mark** for wrong answer. For Section-B, the answer should be rounded off to the nearest integer.

Q.51 The number of N atoms in 681 g of $C_7H_5N_3O_6$ is $x \times 10^{21}$. The value of x is _____.
($N_A = 6.02 \times 10^{23}$ mol^{-1}) (Nearest Integer)

Ans. **[5418.00]**

Sol. Molar mass of $C_7H_5N_3O_6 = 227$ g/mol
681 g of $C_7H_5N_3O_6 = 3$ mol
$\therefore$ 681 g of $C_7H_5N_3O_6$ has 9 mole of N
$= 54.18 \times 10^{23}$ N atoms
$= 5418 \times 10^{21}$

Q.52 The distance between Na^+ and Cl^- ions in solid NaCl of density 43.1 g cm^{-3} is _____ $\times 10^{-10}$ m.
(Nearest Integer) (Given : $N_A = 6.02 \times 10^{23}$ mol^{-1})

Ans. **[1.00]**

Sol. $\rho = \dfrac{Z \times M}{a^3 \times N_A}$

$43.1 = \dfrac{4 \times 58.5}{a^3 \times 6.02 \times 10^{23}}$

$a^3 = 0.9 \times 10^{-23}$
$= 9 \times 10^{-24}$
$a = 2.08 \times 10^{-8}$ cm
$= 2.08 \times 10^{-10}$ m

For NaCl, distance between Na^+ and $Cl^- = \dfrac{a}{2}$

$= 1.04 \times 10^{-10}$ m

Q.53 The longest wavelength of light that can be used for the ionisation of lithium atom (Li) in its ground state is $x \times 10^{-8}$ m. The value of x is _____. (Nearest Integer) (Given : Energy of the electron in the first shell of the hydrogen atom is -2.2×10^{-18} J; $h = 6.63 \times 10^{-34}$ Js and $c = 3 \times 10^{8}$ ms^{-1})

Ans. **[4.00]**

Sol. Energy required for ionisation of Li atom = $2.2 \times 10^{-18} \times \frac{9}{4}$ J [Assume this formule is True for Li atom]

$\because E = \frac{hc}{\lambda}$

$2.2 \times 10^{-18} \times \frac{9}{4} = \frac{6.63 \times 10^{-34} \times 3 \times 10^{8}}{\lambda}$

$\lambda = 4 \times 10^{-8}$ m

Q.54 The standard entropy change for the reaction :
$4Fe(s) + 3O_2(g) \rightarrow 2Fe_2O_3(s)$ is -550 J K^{-1} at 298 K.
[Given: The standard enthalpy change for the reaction is -165 kJ mol^{-1}]. The temperature in K at which the reaction attains equilibrium is _____. (Nearest Integer)

Ans. **[300.00]**

Sol. $\because \Delta S = \frac{\Delta H}{T}$

$\therefore T = \frac{\Delta H}{\Delta S} = \frac{165 \times 10^{3}}{550} = 300$ K

Q.55 1 L aqueous solution of H_2SO_4 contains 0.02 m mol H_2SO_4. 50% of this solution is diluted with deionized water to give 1 L solution (A). In solution (A), 0.01 m mol of H_2SO_4 are added. Total m mols of H_2SO_4 in the final solution is _____ $\times 10^{3}$ m mols.

Ans. **[0.02]**

Sol. Initially one litre contains 0.02 mole

$\therefore$ 50% of this solution will contains 0.01 m mol

After adding 0.01 mol, final solution will contain 0.02 m mol of H_2SO_4 = 0.02 m mol

Correct answer should be 0.02×10^{-3}.

Q.56 The standard free energy change ($\Delta G°$) for 50% dissociation of N_2O_4 into NO_2 at 27°C and 1 atm pressure is $-x$ J mol^{-1}. The value of x is _____. (Nearest Integer)
[Given : R = 8.31 J K^{-1} mol^{-1}, log 1.33 = 0.1239 ln 10 = 2.3]

Ans. **[710.00]**

Sol.

	$N_2O_4 \rightleftharpoons$	$2NO_2$
t = 0	1	0
$t = t_{eq.}$	1 – 0.5	2 × 0.5

$P_{N_2O_4} = 0.33$ atm

$P_{NO_2} = 0.66$ atm

$$K_p = \frac{(P_{NO_2})^2}{P_{N_2O_4}} = \frac{(0.66)^2}{0.33} = 1.33$$

$\because \Delta G = -RT \ln K_p$

$= -8.31 \times 300 \times 2.3 \times \log 1.33 \approx 710 \text{ J mol}^{-1}$

Q.57 In a cell, the following reactions take place

$Fe^{2+} \rightarrow Fe^{3+} + e^-$ $\quad E^\circ_{Fe^{3+}/Fe^{2+}} = 0.77\,V$

$2I^- \rightarrow I_2 + 2e^-$ $\quad E^\circ_{I_2/I^-} = 0.54\,V$

The standard electrode potential for the spontaneous reaction in the cell is $x \times 10^{-2}$ V at 208K. The value of x is _____. (Nearest Integer)

Ans. **[23.00]**

Sol. $E^o_{cell} = E^o_{cathode} - E^o_{anode}$

$= 0.77 - 0.54$

$= 0.23$ V

$= 23 \times 10^{-2}$ V

Q.58 For a given chemical reaction

$\gamma_1 A + \gamma_2 B \rightarrow \gamma_3 C + \gamma_4 D$

Concentration of C changes from 10 m mol dm^{-3} to 20 m mol dm^{-3} in 10 seconds. Rate of appearance of D is 1.5 times the rate of disappearance of B which is twice the rate of disappearance A. The rate of appearance of D has been experimentally determined to be 9 m mol dm^{-3} s^{-1}. Therefore, the rate of reaction is _____ m mol dm^{-3} s^{-1}. (Nearest Integer)

Ans. **[1.00]**

Sol. $\text{Rate} = \frac{1}{r_1}\left(\frac{-d[A]}{dt}\right) = \frac{1}{r_2}\left(\frac{-d[B]}{dt}\right) = \frac{1}{r_3}\left(\frac{-d[C]}{dt}\right) = \frac{1}{r_4}\left(\frac{-d[D]}{dt}\right)$

$$\frac{d[D]}{dt} = \frac{r_4}{r_2}\left(\frac{-d[B]}{dt}\right)$$

$$\frac{r_4}{r_2} = \frac{3}{2}$$

$$\frac{-d[B]}{dt} = \frac{r_2}{r_1}\left(\frac{-d[A]}{dt}\right) \Rightarrow \frac{r_2}{r_1} = 2$$

$r_2 = 2r_1$

$r_4 = 1.5r_2 = 3r_1$

$\frac{d[C]}{dt} = 1$ m.mol dm^{-3} sec^{-1}

$\frac{d[D]}{dt} = 9$ m.mol dm^{-3} sec^{-1}

$\frac{d[D]}{dt} = \frac{r_4}{r_3} \cdot \frac{d[C]}{dt} \Rightarrow \frac{r_4}{r_3} = 9$

$r_4 = 9r_3 = 3r_1$

$\Rightarrow r_1 = 3r_3$

$3r_3A + 6r_3B \rightarrow r_3C + 9r_3D$

$\therefore$ rate of reaction $= \frac{1}{9} \times 9$ m.mol dm^{-3} sec^{-1}

$= 1$ m.mol dm^{-3} sec^{-1}

Q.59 If $[Cu(H_2O)_4]^{2+}$ absorbs a light of wavelength 600 nm for d-d transition, then the value of octahedral crystal field splitting energy for $[Cu(H_2O)_6]^{2+}$ will be_____ $\times 10^{-21}$ J. [Nearest integer]
(Given : $h = 6.63 \times 10^{-34}$ Js and $c = 3.08 \times 10^{8}$ ms^{-1})

Ans. **[765.00]**

Sol. $[Cu(H_2O)_4]^{2+}$ is tetrahedral

$[Cu(H_2O)_6]^{2+}$ is octahedral

$\because \Delta_t = \frac{4}{9} \times \Delta_0$

$\Delta_t = \frac{6.63\times10^{-34}\times3.08\times10^{8}}{600\times10^{-9}}$

$\Delta_0 = \frac{4}{9} \times \frac{6.63\times10^{-34}\times3.08\times10^{8}}{600\times10^{-9}} \approx 765 \times 10^{-21}$ J

Q.60 Number of grams of bromine that will completely react with 5.0 g of pent-1-ene is _____ $\times 10^{-2}$ g. (Atomic mass of Br = 80 g/mol) [Nearest integer]

Ans. **[1143.00]**

Sol.

$CH_3–CH_2–CH_2–CH = CH_2 + Br_2$

$\downarrow$

$CH_3 - CH_2 - CH_2 - \underset{\underset{Br}{|}}{CH} - \overset{\overset{Br}{|}}{CH_2}$

$\frac{5}{70}$ moles of pentene will react with $\frac{5}{70}$ moles of Br_2

$= \frac{5}{70} \times 160$

$= 11.43$ g

$= 1143 \times 10^{-2}$ g

MATHS

Section-A: This section contains 20 multiple choice questions which have only one correct answer. Each question carries **4 marks** for correct answer and **–1 mark** for wrong answer.

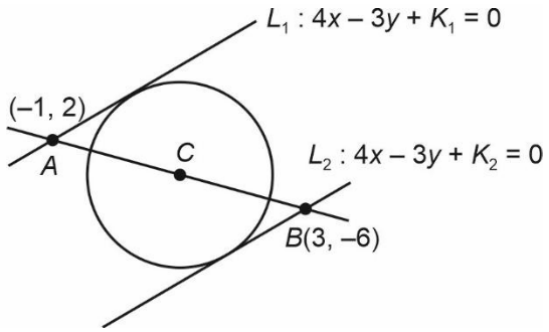
Q.61 Let a circle C touch the lines $L_1 : 4x - 3y + K_1 = 0$ and $L_2 : 4x - 3y + K_2 = 0$, $K_1, K_2 \in \mathbf{R}$. If a line passing through the centre of the circle C intersects L_1 at (–1, 2) and L_2 at (3, –6), then the equation of the circle C is :

(1) $(x-1)^2 + (y-2)^2 = 4$ (2) $(x+1)^2 + (y-2)^2 = 4$

(3) $(x-1)^2 + (y+2)^2 = 16$ (4) $(x-1)^2 + (y-2)^2 = 16$

Ans. **[3]**

Sol.



Co-ordinate of centre $C \equiv \left(\frac{3+(-1)}{2}, \frac{-6+2}{2}\right) \equiv (1, -2)$

L_1 is passing through A

$\Rightarrow -4 - 6 + K_1 = 0$

$\Rightarrow K_1 = 10$

L_2 is passing through B

$\Rightarrow 12 + 18 + K_2 = 0$

$\Rightarrow K_2 = -30$

Equation of $L_1 : 4x - 3y + 10 = 0$

Equation of $L_2 : 4x - 3y - 30 = 0$

Diameter of circle $= \left|\frac{10+30}{\sqrt{4^2+(-3)^2}}\right| = 8$

$\Rightarrow$ Radius = 4

Equation of circle $(x-1)^2 + (y+2)^2 = 16$

Q.62 The value of $\int_0^{\pi} \frac{e^{\cos x} \sin x}{(1+\cos^2 x)(e^{\cos x} + e^{-\cos x})} dx$ is equal to :

(1) $\frac{\pi^2}{4}$ (2) $\frac{\pi^2}{2}$ (3) $\frac{\pi}{4}$ (4) $\frac{\pi}{2}$

Ans. **[3]**

Sol. $\int_0^{\pi} \frac{e^{\cos x} \sin x}{(1+\cos^2 x)(e^{\cos x} + e^{-\cos x})} dx$

CAREER POINT

Let cos x = t

sin x dx = dt

$$= \int_{1}^{-1} \frac{-e^t dt}{(1+t^2)(e^t+e^{-t})}$$

$$I = \int_{-1}^{1} \frac{e^t}{(1+t^2)(e^t+e^{-t})} dt \quad ...(i)$$

$$I = \int_{-1}^{1} \frac{e^{-t}}{(1+t^2)(e^{-t}+e^t)} dt \quad ...(ii)$$

Adding (i) and (ii)

$$2I = \int_{-1}^{1} \frac{dt}{1+t^2}$$

$$2I = \tan^{-t}\Big]_{-1}^{1}$$

$$2I = \frac{\pi}{4} - \left(-\frac{\pi}{4}\right)$$

$$2I = \frac{\pi}{2}$$

$$I = \frac{\pi}{4}$$

Q.63 Let a, b and c be the length of sides of a triangle ABC such that $\frac{a+b}{7} = \frac{b+c}{8} = \frac{c+a}{9}$. If r and R are the radius of incircle and radius of circumcircle of the triangle ABC, respectively, then the value of $\frac{R}{r}$ is equal to :

(1) $\frac{5}{2}$ (2) 2 (3) $\frac{3}{2}$ (4) 1

Ans. **[1]**

Sol. $\frac{a+b}{7} = \frac{b+c}{8} = \frac{c+a}{9} = \lambda$

a + b = 7λ

b + c = 8λ

c + a = 9λ

a + b + c = 12λ

∴ a = 4λ, b = 3λ, c = 5λ

$$S = \frac{4\lambda+3\lambda+5\lambda}{2} = 6\lambda$$

$$\Delta = \sqrt{S(s-a)(s-b)(s-c)} = \sqrt{(6\lambda)(2\lambda)(3\lambda)(\lambda)} = 6\lambda^2$$

$$R = \frac{abc}{4\Delta} = \frac{(4\lambda)(3\lambda)(5\lambda)}{4(6\lambda^2)} = \frac{5}{2}\lambda$$

$$r = \frac{\Delta}{s} = \frac{6\lambda^2}{6\lambda} = \lambda$$

$$\frac{R}{r} = \frac{\frac{5}{2}\lambda}{\lambda} = \frac{5}{2}$$

Q.64 Let $f : \mathbf{N} \to \mathbf{R}$ be a function such that $f(x + y) = 2f(x)\, f(y)$ for natural numbers x and y. If $f(1) = 2$, then the value of α for which $\sum_{k=1}^{10} f(\alpha + k) = \frac{512}{3}(2^{20} - 1)$ holds, is :

(1) 2 (2) 3 (3) 4 (4) 6

Ans. **[3]**

Sol. $f(x + y) = 2f(x)\, f(y)$ & $f(1) = 2$

$x = y = 1$

$$\left.\begin{array}{l} \Rightarrow f(2) = 2^3 \\ x = 2, y = 1 \\ \Rightarrow f(3) = 2^5 \end{array}\right\} f(x) = 2^{(2x-1)}$$

Now,

$$\sum_{k=1}^{10} f(\alpha + k) = \frac{512}{3}(2^{20} - 1)$$

$$2\sum_{k=1}^{10} f(\alpha) f(k) = \frac{512}{3}(2^{20} - 1)$$

$$2f(\alpha)\,[f(1) + f(2) + ... + f(10)] = \frac{512}{3}(2^{20} - 1)$$

$$= 2f(\alpha)\,[2 + 2^3 + 2^5 + ... \text{ upto 10 terms}] = \frac{512}{3}(2^{20} - 1)$$

$$= 2f(\alpha) \cdot 2\left(\frac{2^{20} - 1}{4 - 1}\right) = \frac{512}{3}(2^{20} - 1)$$

$f(\alpha) = 128 = 2^{2\alpha} - 1$

$= 2\alpha - 1 = 7$

$\Rightarrow \alpha = 4$

Q.65 Let A be a 3×3 real matrix such that

$$A\begin{pmatrix}1\\1\\0\end{pmatrix} = \begin{pmatrix}1\\1\\0\end{pmatrix};\ A\begin{pmatrix}1\\0\\1\end{pmatrix} = \begin{pmatrix}-1\\0\\1\end{pmatrix} \text{ and } A\begin{pmatrix}0\\0\\1\end{pmatrix} = \begin{pmatrix}1\\1\\2\end{pmatrix}$$

If $X = (x_1, x_2, x_3)^T$ and I is an identity matrix of order 3, then the system $(A - 2I)X = \begin{pmatrix}4\\1\\1\end{pmatrix}$ has :

(1) no solution (2) Infinitely many solutions

(3) Unique solution (4) Exactly two solutions

Ans. **[2]**

Sol. Let $A = \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix}$

$$A\begin{bmatrix}1\\1\\0\end{bmatrix}=\begin{bmatrix}1\\1\\0\end{bmatrix}\Rightarrow\begin{bmatrix}a&b&c\\d&e&f\\g&h&i\end{bmatrix}\begin{bmatrix}1\\1\\0\end{bmatrix}=\begin{bmatrix}1\\1\\0\end{bmatrix}\Rightarrow\begin{matrix}a+b=1\\d+e=1\\g+h=0\end{matrix}$$

$$A\begin{bmatrix}1\\0\\1\end{bmatrix}=\begin{bmatrix}-1\\0\\1\end{bmatrix}\Rightarrow\begin{bmatrix}a&b&c\\d&e&f\\g&h&i\end{bmatrix}\begin{bmatrix}1\\0\\1\end{bmatrix}=\begin{bmatrix}-1\\0\\1\end{bmatrix}\Rightarrow\begin{matrix}a+c=-1\\d+f=0\\g+i=1\end{matrix}$$

$$A\begin{bmatrix}0\\0\\1\end{bmatrix}=\begin{bmatrix}1\\1\\2\end{bmatrix}\Rightarrow\begin{bmatrix}a&b&c\\d&e&f\\g&h&i\end{bmatrix}\begin{bmatrix}0\\0\\1\end{bmatrix}=\begin{bmatrix}1\\1\\2\end{bmatrix}\Rightarrow\begin{matrix}c=1\\f=1\\i=2\end{matrix}$$

Solving will get

$a = -2, b = 3, c = 1, d = -1, e = 2, f = 1, g = -1, h = 1, i = 2$

$$A=\begin{bmatrix}-2&3&1\\-1&2&1\\-1&1&2\end{bmatrix}\Rightarrow A=2I=\begin{bmatrix}-4&3&1\\-1&0&1\\-1&1&0\end{bmatrix}$$

$$(A-2I)x=\begin{bmatrix}4\\1\\1\end{bmatrix}$$

$\Rightarrow -4x_1 + 3x_2 + x_3 = 4$ ...(i)

$-x_1 + x_3 = 1$ ...(ii)

$-x_1 + x_2 = 1$ ...(iii)

So 3(iii) + (ii) = (i)

$\therefore$ Infinite solution

Q.66 Let $f : \mathbf{R} \to \mathbf{R}$ be defined as

$f(x) = x^3 + x - 5$

If g(x) is a function such that $f(g(x)) = x, \forall$ 'x' $\in \mathbf{R}$, then g'(63) is equal to_____.

(1) $\frac{1}{49}$ (2) $\frac{3}{49}$ (3) $\frac{43}{49}$ (4) $\frac{91}{49}$

Ans. **[1]**

Sol. $f(x) = 3x^2 + 1$

$f'(x)$ is bijective function and $f(g(x)) = x \Rightarrow g(x)$ is inverse of $f(x)$

$g(f(x)) = x$

$g'(f(x)).f'(x) = 1$

$$g'(f(x)) = \frac{1}{3x^2+1}$$

Put x = 4 we get

$$g'(63) = \frac{1}{49}$$

CAREER POINT

Q.67 Consider the following two propositions :

P1 : $\sim (p \rightarrow \sim q)$

P2 : $(p \wedge \sim q) \wedge ((\sim p) \vee q)$

If the proposition $p \rightarrow ((\sim p) \vee q)$ is evaluated as FALSE, then :

(1) P1 is TRUE and P2 is FALSE (2) P1 is FALSE and P2 is TRUE

(3) Both P1 and P2 are FALSE (4) Both P1 and P2 are TRUE

Ans. **[3]**

Sol. Given $p \rightarrow (\sim p \vee q)$ is false

$\Rightarrow \sim p \vee q$ is false and p is true

Now p = true.

$\sim T \vee q = F$

$F \vee q = F \Rightarrow q$ is false

P1 : $\sim (T \rightarrow \sim F) \equiv \sim (T \rightarrow T) \equiv$ False.

P2 : $(T \wedge \sim F) \wedge (\sim T \vee F) \equiv (T \wedge T) \wedge (F \vee F) \equiv T \wedge F \equiv$ False.

Q.68 If $\frac{1}{2\cdot 3^{10}} + \frac{1}{2^2\cdot 3^9} + ... + \frac{1}{2^{10}\cdot 3} = \frac{K}{2^{10}\cdot 3^{10}}$, then the remainder when K is divided by 6 is -

(1) 1 (2) 2 (3) 3 (4) 5

Ans. **[4]**

Sol. $\frac{1}{2\cdot 3^{10}} + \frac{1}{2^2\cdot 3^9} + ... + \frac{1}{2^{10}\cdot 3} = \frac{K}{2^{10}\cdot 3^{10}}$

$$\Rightarrow \frac{1}{2\cdot 3^{10}}\left[\frac{\left(\frac{3}{2}\right)^{10}-1}{\frac{3}{2}-1}\right] = \frac{K}{2^{10}\cdot 3^{10}}$$

$$= \frac{3^{10}-2^{10}}{2^{10}\cdot 3^{10}} = \frac{K}{2^{10}\cdot 3^{10}} \Rightarrow K = 3^{10} - 2^{10}$$

Now $K = (1+2)^{10} - 2^{10}$

$= {}^{10}C_0 + {}^{10}C_1 2 + {}^{10}C_2 2^3 + ... + {}^{10}C_{10} 2^{10} - 2^{10}$

$= {}^{10}C_0 + {}^{10}C_1 2 + 6\lambda + {}^{10}C_9 \cdot 2^9$

$= 1 + 20 + 5120 + 6\lambda$

$= 5136 + 6\lambda + 5$

$= 6\mu + 5$

$\lambda, \mu \in N$

$\therefore$ remainder = 5

Q.69 Let $f(x)$ be a polynomial function such that $f(x) + f'(x) + f''(x) = x^5 + 64$. Then, the value of $\lim\limits_{x\to 1}\frac{f(x)}{x-1}$ is equal to :

(1) – 15 (2) – 60 (3) 60 (4) 15

Ans. **[1]**

Sol. $\lim\limits_{x\to 1}\dfrac{f(x)}{x-1}$

$f(x) + f'(x) + f''(x) = x^5 + 64$

Let $f(x) = x^5 + ax^4 + bx^3 + cx^2 + dx + e$

$f'(x) = 5x^4 + 4ax^3 + 3bx^2 + 2cx + d$

$f''(x) = 20x^3 + 12ax^2 + 6bx + 2c$

$x^5 + (a+5)x^4 + (b + 4a + 20)x^3 + (c + 3b + 12a)x^2 + (d + 2c + 6b)x + e + d + 2c = x^5 + 64$

$\Rightarrow a + 5 = 0$

$b + 4a + 20 = 0$

$c + 3b + 12a = 0$

$d + 2c + 6b = 0$

$e + d + 2c = 64$

$\therefore a = -5, b = 0, c = 60, d = -120, e = 64$

$\therefore f(x) = x^5 - 5x^4 + 60x^2 - 120x + 64$

Now, $\lim\limits_{x\to 1}\dfrac{x^5 - 5x^4 + 60x^2 - 120x + 64}{x-1}$ is $\left(\dfrac{0}{0}\text{ from}\right)$

By L′ Hospital rule

$\lim\limits_{x\to 1}\dfrac{5x^4 - 20x^3 + 120x - 120}{1} = -15$

Q.70 Let E_1 and E_2 be two events such that the conditional probabilities $P(E_1 \mid E_2) = \dfrac{1}{2}$, $P(E_2 \mid E_1) = \dfrac{3}{4}$ and $P(E_1 \cap E_2) = \dfrac{1}{8}$. Then -

(1) $P(E_1 \cap E_2) = P(E_1) \cdot P(E_2)$

(2) $P(E'_1 \cap E'_2) = P(E'_1) \cdot P(E_2)$

(3) $P(E_1 \cap E'_2) = P(E_1) \cdot P(E_2)$

(4) $P(E'_1 \cap E_2) = P(E_1) \cdot P(E_2)$

Ans. **[3]**

Sol. $P\left(\dfrac{E_1}{E_2}\right) = \dfrac{1}{2} \Rightarrow \dfrac{P(E_1 \cap E_2)}{P(E_2)} = \dfrac{1}{2}$

$P\left(\dfrac{E_2}{E_1}\right) = \dfrac{3}{4} \Rightarrow \dfrac{P(E_2 \cap E_1)}{P(E_1)} = \dfrac{3}{4}$

$P(E_1 \cap E_2) = \dfrac{1}{8}$

$P(E_2) = \dfrac{1}{4}$, $P(E_1) = \dfrac{1}{6}$

(1) $P(E_1 \cap E_2) = \dfrac{1}{8}$ and $P(E_1)\cdot P(E_2) = \dfrac{1}{24}$

$\Rightarrow P(E_1 \cap E_2) \neq P(E_1)\cdot P(E_2)$

(2) $P(E'_1 \cap E'_2) = 1 - P(E_1 \cup E_2)$

$= 1 - \left[\dfrac{1}{4} + \dfrac{1}{6} - \dfrac{1}{8}\right] = \dfrac{17}{24}$

$P(E'_1) = \dfrac{3}{4} \Rightarrow P(E'_1)\, P(E_2) = \dfrac{3}{24}$

$\Rightarrow P(E'_1 \cap E'_2) \neq P(E'_1) \cdot P(E_2)$

(3) $P(E_1 \cap E'_2) = P(E_1) - P(E_1 \cap E_2)$

$= \frac{1}{6} - \frac{1}{8} = \frac{1}{24}$

$P(E_1) \cdot P(E_2) = \frac{1}{24}$

$\Rightarrow P(E_1 \cap E'_2) = P(E_1) \cdot P(E_2)$

(4) $P(E'_1 \cap E_2) = P(E_2) - P(E_1 \cap E_2)$

$= \frac{1}{4} - \frac{1}{8} = \frac{1}{8}$

$P(E_1)\, P(E_2) = \frac{1}{24}$

$\Rightarrow P(E'_1 \cap E_2) \neq P(E_1) \cdot P(E_2)$

Q.71 Let $A = \begin{bmatrix} 0 & -2 \\ 2 & 0 \end{bmatrix}$. If M and N are two matrices given by $M = \sum_{k=1}^{10} A^{2k}$ and $N = \sum_{k=1}^{10} A^{2k-1}$ then MN^2 is :

(1) a non-identity symmetric matrix (2) a skew-symmetric matrix

(3) neither symmetric nor skew-symmetric matrix (4) an identity matrix

Ans. **[1]**

Sol. $A = \begin{bmatrix} 0 & -2 \\ 2 & 0 \end{bmatrix}$

$A^2 = \begin{bmatrix} 0 & -2 \\ 2 & 0 \end{bmatrix} \begin{bmatrix} 0 & -2 \\ 2 & 0 \end{bmatrix} = \begin{bmatrix} -4 & 0 \\ 0 & -4 \end{bmatrix} = -4I$

$M = A^2 + A^4 + A^6 + ... + A^{20}$

$= -4I + 16I - 64I + ...$ upto 10 terms

$= -I[4 - 16 + 64 ... +$ upto 10 terms$]$

$= -I.4\left[\frac{(-4)^{10} - 1}{-4-1}\right] = \frac{4}{5}(2^{20} - 1)I$

$N = A^1 + A^3 + A^5 + ... + A^{19}$

$= A - 4A + 16A + ...$ upto 10 terms

$= A\left(\frac{(-4)^{10} - 1}{-4-1}\right) = -\left(\frac{2^{10} - 1}{5}\right)A$

$N^2 = \frac{(2^{10} - 1)^2}{2^5} A^2 = \frac{-4}{25}(2^{20} - 1)^2 I$

$MN^2 = \frac{-16}{125}(2^{20} - 1)^3 I = KI \qquad (K \neq \pm 1)$

$(MN^2)^T = (KI)^T = KI$

$\therefore$ A is correct

Q.72 Let g : (0, ∞) → R be a differentiable function such that $\int\left(\frac{x(\cos x-\sin x)}{e^x+1}+\frac{g(x)(e^x+1-xe^x)}{(e^x+1)^2}\right)dx=\frac{xg(x)}{e^x+1}+c$,

for all x > 0. where c is and arbitrary constant. then :

(1) g is decreasing in $\left(0,\frac{\pi}{4}\right)$

(2) g′ is increasing in $\left(0,\frac{\pi}{4}\right)$

(3) g + g′ is increasing in $\left(0,\frac{\pi}{2}\right)$

(4) g – g′ is increasing in $\left(0,\frac{\pi}{2}\right)$

Ans. **[4]**

Sol. $\int\left(\frac{x(\cos x-\sin x)}{e^x+1}+\frac{g(x)(e^x+1-xe^x)}{(e^x+1)^2}\right)dx=\frac{xg(x)}{e^x+1}+c$

Differentiating on both sides

$$\frac{x(\cos x-\sin x)}{e^x+1}+\frac{g(x)(e^x+1-xe^x)}{(e^x+1)^2}=\frac{(e^x+1)\left(g(x)+xg^{\frac{1}{x}}\right)-xg(x)e^x}{(e^x+1)^2}$$

$$=\frac{g(x)[e^x+1-xe^x]}{(e^x+1)^2}+\frac{xg'(x)(e^x+1)}{(e^x+1)^2}$$

$$=\frac{x(\cos x-\sin x)}{e^x+1}=\frac{xg'(x)}{e^x+1}$$

$\Rightarrow g'(x) = \cos x - \sin x > 0$ in $\left(0,\frac{\pi}{4}\right)$

$\Rightarrow$ g(x) is increasing in $\left(0,\frac{\pi}{4}\right)$ $\Rightarrow$ (1) is wrong

Now, $g''(x) = -\sin x - \cos x < 0$ in $\left(0,\frac{\pi}{4}\right)$

$\Rightarrow$ g(x) is increasing in $\left(0,\frac{\pi}{4}\right)$ $\Rightarrow$ (2) is wrong

Let h(x) = g(x) + g′(x)

$\Rightarrow h'(x) = g'(x) + g''(x) = -2\sin x < 0$ in $x \in \left(0,\frac{\pi}{2}\right)$

$\Rightarrow$ g + g′ is decreasing in $\left(0,\frac{\pi}{2}\right)$ $\Rightarrow$ (3) is wrong

Let J(x) = g(x) – g′(x)

$J'(x) = g'(x) - g''(x) = 2\cos x > 0$ in $\left(0,\frac{\pi}{2}\right)$

$\Rightarrow$ g – g′ is increasing in $\left(0,\frac{\pi}{2}\right)$ $\Rightarrow$ correct

Q.73 Let $f : R \to R$ and $g : R \to R$ be two functions defined by $f(x) = \log_e (x^2 + 1) - e^{-x} + 1$ and $g(x) = \frac{1 - 2e^{2x}}{e^x}$.

Then, for which of the following range of α, the inequality $f\left(g\left(\frac{(\alpha-1)^2}{3}\right)\right) > f\left(g\left(\alpha - \frac{5}{3}\right)\right)$ holds ?

(1) (2, 3) (2) (–2, –1) (3) (1, 2) (4) (–1, 1)

Ans. **[1]**

Sol. $f(x) = \log_e (x^2 + 1) - e^{-x} + 1$

$f'(x) = \frac{2x}{x^2+1} + e^{-x}$

$= \frac{2}{x + \frac{1}{x}} + e^{-x} > 0 \ \forall x \in R$

$g(x) = e^{-x} - 2e^x$

$g'(x) = -e^{-x} - 2e^x < 0 \qquad \forall x \in R$

$\Rightarrow f(x)$ is increasing and $g(x)$ is decreasing function.

$f\left(g\left(\frac{(\alpha-1)^2}{3}\right)\right) > f\left(g\left(\alpha - \frac{5}{3}\right)\right)$

$\Rightarrow \frac{(\alpha-1)^2}{3} < \alpha - \frac{5}{3}$

$= \alpha^2 - 5\alpha + 6 < 0$

$= (\alpha - 2)(\alpha - 3) < 0$

$= \alpha \in (2, 3)$

Q.74 Let $\vec{a} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$ $a_i > 0$, $i = 1, 2, 3$ be vector which makes equal angles with the coordinate axes OX, OY and OZ. Also, let the projection of $\vec{a}$ on the vector $3\hat{i} + 4\hat{j}$ be 7. Let $\vec{b}$ be a vector obtained by rotating $\vec{a}$ with 90°. If $\vec{a}$, $\vec{b}$ and x-axis are coplanar, then projection of a vector $\vec{b}$ on $3\hat{i} + 4\hat{j}$ is equal to :

(1) $\sqrt{7}$ (2) $\sqrt{2}$ (3) 2 (4) 4

Ans. **[2]**

Sol. $\cos^2\alpha + \cos^2\beta + \cos^2\gamma = 1 \Rightarrow \cos^2\alpha = \frac{1}{3} \Rightarrow \cos\alpha = \frac{1}{\sqrt{3}}$

$\vec{a} = \frac{\lambda}{3}(\hat{i} + \hat{j} + \hat{k}), \lambda > 0$

$\frac{\lambda}{\sqrt{3}} \frac{(\hat{i} + \hat{j} + \hat{k}).(3\hat{i} + 4\hat{j})}{\sqrt{3^2 + 4^2}} = 7$

$\Rightarrow \frac{\lambda}{\sqrt{3}}(3 + 4) = 7 \times 5$

$\therefore \lambda = 5\sqrt{3}$

$\vec{a} = 5(\hat{i} + \hat{j} + \hat{k})$

Let $\vec{b} = p\hat{i} + q\hat{j} + r\hat{k}$

$\vec{a} \cdot \vec{b} = 0$ and $[\vec{a}\ \vec{b}\ \hat{i}] = 0$

$\Rightarrow p + q + r = 0$ ...(i)

& $\begin{vmatrix} p & q & r \\ 1 & 1 & 1 \\ 1 & 0 & 0 \end{vmatrix} = 0 \Rightarrow$ $\boxed{q = r}$ $\boxed{p = -2r}$

$\vec{b} = -2r\hat{i} + r\hat{j} + r\hat{k}$

$\vec{b} = r(-2\hat{i} + \hat{j} + \hat{k})$

Now $|\vec{a}| = |\vec{b}|$

$5\sqrt{3} = |r|\sqrt{b} \Rightarrow |r| = \dfrac{5}{\sqrt{2}}$

$\Rightarrow$ Projection of $\vec{b}$ on $3\hat{i} + 4\hat{j} = \left|\dfrac{\vec{b}.(3\hat{i} + 4\hat{j})}{\sqrt{3^2 + 4^2}}\right| = |r|\dfrac{(-6+4)}{5} = \left|\dfrac{-2r}{5}\right|$

Projection $= \dfrac{2}{5} \times \dfrac{5}{\sqrt{2}} = \sqrt{2}$

$\therefore$ Option (2) is correct

Q.75 Let y = y(x) be the solution of the differential equation $(x + 1)y' - y = e^{3x}(x + 1)^2$, with $y(0) = \dfrac{1}{3}$. Then, the point $x = -\dfrac{4}{3}$ for the curve y = y(x) is -

(1) not a critical point
(2) a point of local minima
(3) a point of local maxima
(4) a point of inflection

Ans. **[2]**

Sol. $(x + 1)\dfrac{dy}{dx} - y = e^{3x}(x + 1)^2$

$\dfrac{dy}{dx} - \dfrac{y}{x+1} = e^{3x}(x + 1)$

If $e^{-\int \frac{1}{x+1}dx} = e^{-\log(x+1)} = \dfrac{1}{x+1}$

$\therefore y\left(\dfrac{1}{x+1}\right) = \int \dfrac{e^{3x}(x+1)}{x+1}dx$

$\dfrac{y}{x+1} = \int e^{3x}dx$

$\boxed{\dfrac{y}{x+1} = \dfrac{e^{3x}}{3} + c}$

$\because y(0) = \dfrac{1}{3}$

$\frac{1}{3} = \frac{1}{3} + c$

$\therefore\ c = 0$

So $\boxed{y = \frac{e^{3x}}{3}(x+1)}$

$y' = e^{3x}(x+1) + \frac{e^{3x}}{3} = e^{3x}\left(x + \frac{4}{3}\right)$

$y'' = 3e^{3x}\left(x + \frac{4}{3}\right) + e^{3x} = e^{3x}(3x+5)$

$y' = 0$ at $x = \frac{-4}{3}$ & $y'' = e^{-4}(1) > 0$ at $x = \frac{-4}{3}$

$\Rightarrow x = \frac{-4}{3}$ is point of local minima

Q.76 If $y = m_1x + c_1$ and $y = m_2x + c_2$, $m_1 \neq m_2$ are two common tangents of circle $x^2 + y^2 = 2$ and parabola $y^2 = x$, then the value of $8|m_1m_2|$ is equal to :

(1) $3 + 4\sqrt{2}$ (2) $-5 + 6\sqrt{2}$ (3) $-4 + 3\sqrt{2}$ (4) $7 + 6\sqrt{2}$

Ans. **[3]**

Sol. Let tangent to $y^2 = x$ be

$y = mx + \frac{1}{4m}$

For it being tangent to circle

$\left|\frac{\frac{1}{4}m}{\sqrt{1+m^2}}\right| = \sqrt{2}$

$\Rightarrow 32m^4 + 32m^2 - 1 = 0$

$\Rightarrow m^2 = \frac{-32 \pm \sqrt{(32)^2 + 4(32)}}{64}$

$\Rightarrow 8m_1m_2 = -4 + 3\sqrt{2}$

Q.77 Let Q be the mirror image of the point P(1, 0, 1) with respect to the plane S: $x + y + z = 5$. If a line L passing through $(1, -1, -1)$, parallel to the line PQ meets the plane S at R, then QR^2 is equal to :

(1) 2 (2) 5 (3) 7 (4) 11

Ans. **[2]**

Sol. As L is parallel to PQ d.r.s of S is < 1, 1, 1 >

$\therefore\ L = \frac{x-1}{1} = \frac{y+1}{1} = \frac{z+1}{1}$

Point of intersection of L and S be λ

$\Rightarrow (\lambda + 1) + (\lambda - 1) + (\lambda - 1) = S$

$\Rightarrow \lambda = 2$

$\therefore$ R = (3, 1, 1)

Let Q (α, β, γ)

$\Rightarrow \frac{\alpha-1}{1}=\frac{\beta}{1}=\frac{\gamma-1}{1}=\frac{-2(-3)}{3}$

$\Rightarrow \alpha = 3, \beta = 2, \gamma = 3$

$\Rightarrow Q \equiv (3,2,3)$

$(QR)^2 = 0^2 + (1)^2 + (2)^2 = 5$

Q.78 If the solution curve y = y(x) of the differential equation $y^2dx + (x^2 - xy + y^2)dy = 0$, which passes through the point (1,1) and intersects the line $y = \sqrt{3}\,x$ at the point $\left(\alpha, \sqrt{3}\,\alpha\right)$, then value of $\log_e\left(\sqrt{3}\,\alpha\right)$ is equal to :

(1) $\frac{\pi}{3}$ (2) $\frac{\pi}{2}$ (3) $\frac{\pi}{12}$ (4) $\frac{\pi}{6}$

Ans. **[3]**

Sol. $\frac{dy}{dx}=\frac{y^2}{xy-x^2-y^2}$

Put y = vx we get

$v + x\frac{dv}{dx}=\frac{v^2}{v-1-v^2}$

$\Rightarrow x\frac{dv}{dx}=\frac{v^2-v^2+v+v^3}{v-1-v^2}$

$\Rightarrow \int\frac{v-1-v^2}{v(1+v^2)}dv=\int\frac{dx}{x}$

$\tan^{-1}\left(\frac{y}{x}\right)-\ln\left(\frac{y}{x}\right)=\ln x + c$

As it passes through (1, 1)

$c = \frac{\pi}{4}$

$\Rightarrow \tan^{-1}\left(\frac{y}{x}\right)-\ln\left(\frac{y}{x}\right)=\ln x+\frac{\pi}{4}$

Put $y = \sqrt{3}\,x$ we get

$\Rightarrow \frac{\pi}{3}-\ln\sqrt{3}=\ln x+\frac{\pi}{4}$

$\Rightarrow \ln x=\frac{\pi}{12}-\ln\sqrt{3}=\ln\alpha$

$\therefore \ln(\sqrt{3}\alpha)=\ln\sqrt{3}+\ln\alpha$

$=\ln\sqrt{3}+\frac{\pi}{12}-\ln\sqrt{3}=\frac{\pi}{12}$

CAREER POINT

Q.79 Let x = 2t, $y = \frac{t^2}{3}$ be a conic. Let S be the focus and B be the point on the axis of the conic such that SA ⊥ BA, where A is any point on the conic. If k is the ordinate of the centroid of the ΔSAB, then $\lim_{t\to 1} k$ is equal to

(1) $\frac{17}{18}$ (2) $\frac{19}{18}$ (3) $\frac{11}{18}$ (4) $\frac{13}{18}$

Ans. **[4]**

Sol. $x = 2t, y = \frac{2}{3}$

$t \to 1$ A≡ $\left(2, \frac{1}{3}\right)$

Given conic is $x^2 = 12y \Rightarrow S \equiv (0,3)$

Let $B \equiv (0, \beta)$

Given SA ⊥ BA

$$\left(\frac{\frac{1}{3}}{2-3}\right)\left(\frac{\beta-\frac{1}{3}}{-2}\right) = -1$$

$$\Rightarrow \left(\beta-\frac{1}{3}\right)\frac{1}{3} = -2$$

$$\Rightarrow \beta = \frac{1}{3}\left(\frac{-17}{3}\right)$$

Ordinate of centroid = $k = \frac{\beta+\frac{1}{3}+3}{3}$ (as $t \to 1$)

$$= \frac{\frac{-17}{9}+\frac{10}{3}}{3} = \frac{13}{18}$$

Q.80 Let a circle C in complex plane pass through the points $z_1 = 3 + 4i$, $z_2 = 4 + 3i$ and $z_3 = 5i$. If $z(\neq z_1)$ is a point on C such that the line through z and z_1 is perpendicular to the line through z_2 and z_3, then arg(z) is equal to:

(1) $\tan^{-1}\left(\frac{2}{\sqrt{5}}\right) - \pi$ (2) $\tan^{-1}\left(\frac{24}{7}\right) - \pi$ (3) $\tan^{-1}(3) - \pi$ (4) $\tan^{-1}\left(\frac{3}{4}\right) - \pi$

Ans. **[2]**

Sol. $z_1 = 3 + 4i$, $z_2 = 4 + 3i$ and $z_3 = 5i$

Clearly $C \equiv x^2 + y^2 = 25$

Let z (x, y)

$$\Rightarrow \left(\frac{y-4}{x-3}\right)\left(\frac{2}{-4}\right) = -1$$

$\Rightarrow y = 2x - 2 = L$

∴ z is intersection of C & L

$$\Rightarrow z = \left(\frac{-7}{5}, \frac{-24}{5}\right)$$

$$\therefore \text{Arg}(z) = -\pi + \tan^{-1}\left(\frac{24}{7}\right)$$

CAREER POINT

Section-B: This section contains 10 questions. In Section-B, attempt any **five questions out of 10.** The answer to each of the questions is a numerical value. For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the second decimal place; e.g. 06.25, 07.00, –00.33, –00.30, 30.27, –27.30). Each question carries **4 marks** for correct answer and **–1 mark** for wrong answer.

Q.81 Let C_r denote the binomial coefficient of x^r in the expansion of $(1 + x)^{10}$. If for $\alpha, \beta \in R$, $C_1 + 3.2\, C_2 + 5.3\, C_3 + \ldots$. upto to terms $= \dfrac{\alpha \times 2^{11}}{2^\beta - 1}\left(C_0 + \dfrac{C_1}{2} + \dfrac{C_2}{3} + \ldots\ldots \text{upto} 10 \text{ terms}\right)$ then the value of $\alpha + \beta$ is equal to _____.

Ans. **[286.00]**

Sol. Given than $C_1 + 2.3C_2 + 5.3C_3 + \ldots.10 \text{ terms} = \dfrac{\alpha.2^{11}}{2^\beta - 1}\left(C_1 + \dfrac{C_2}{2} + \ldots.\right)$

$$\Rightarrow \sum_{r=1}^{10} r(2r-1)C_r = \frac{\alpha.2^{11}}{2^\beta - 1}\left(\sum_{r=1}^{10} \frac{C_r}{r}\right)$$

Using $C_1 + 2C_2 + \ldots\ldots. nC_n = n.2^{n-1}$,

$1^2C_1 + 2^2C_2 + \ldots.. + n^2C_n = n.2^{n-1} + n(n-1)2^{n-2}$

and $C_0 + \dfrac{C_1}{2} + \ldots.. \dfrac{C_n}{n+1} = \dfrac{2^{n+1} - 1}{n+1}$ we get

$$\Rightarrow 2(10.2^9 + 10.9.2^8) - 10.2^9 = \frac{\alpha.2^{11}}{2^\beta - 1}\ \frac{(2^{11} - 1)}{11}$$

Comparing both side we get

$$2^{11}.25 = \frac{\alpha.2^{11}}{2^\beta - 1}\frac{(2^{11} - 1)}{11}$$

$\Rightarrow \alpha = 25 \times 11 = 275$ & $\beta = 11$

$\Rightarrow \alpha + \beta = 286$

Q.82 The number of 3-digit Odd numbers, whose sum of digits is a multiple of 7, is _____.

Ans. **[63.00]**

Sol. For odd number unit place shall be 1, 3, 5, 7 or 9.

$\therefore$ x y 1, x y 3, x y 5, x y 7, x y 9 are the type of numbers.

If x y 1 then

x + y = 6, 13, 20 … Cases are required

i.e., 6 + 6 + 0 + … = 12 ways

If x y 3 then

x + y = 4, 11, 18, …. Cases are required

i.e., 4 + 8 + 1 + 0 … = 13 ways

Similarly for x y 5, we have

x + y = 2, 9, 16, …

i.e., 2 + 9 + 3 = 14 ways

for x y 7 we have

x + y = 0, 7, 14, ….

i.e., 0 + 7 + 5 = 12 ways

And for x y 9 we have

x + y = 5, 12, 19 …

i.e., 5 + 7 + 0 … = 12 ways

$\therefore$ Total 63 ways

Q.83 Let θ be the angle between the vectors $\vec{a}$ and $\vec{b}$, where $|\vec{a}| = 4$, $|\vec{b}| = 3$ and $\theta \in \left(\frac{\pi}{4}, \frac{\pi}{3}\right)$. Then $\left|(\vec{a}-\vec{b})\times(\vec{a}+\vec{b})\right|^2 + 4\,(\vec{a}\cdot\vec{b})^2$ is equal to _____.

Ans. **[576.00]**

Sol. $|(\vec{a}-\vec{b})\times(\vec{a}+\vec{b})|^2 + 4(\vec{a}\cdot\vec{b})^2$

$\Rightarrow |\vec{a}\times\vec{a} + \vec{a}\times\vec{b} - \vec{b}\times\vec{a} - \vec{b}\times\vec{b}|^2 + 4(\vec{a}\cdot\vec{b})^2$

$\Rightarrow |2(\vec{a}\times\vec{b})|^2 + 4(\vec{a}\cdot\vec{b})^2$

$\Rightarrow 4(\vec{a}\times\vec{b})^2 + (\vec{a}\cdot\vec{b})^2$

$\Rightarrow 4|\vec{a}|^2|\vec{b}|^2 = 4.16.9 = 576$

Q.84 Let the abscissae of the two points P and Q be the roots of $2x^2 - rx + p = 0$ and the ordinates of P and Q be the roots of $x^2 - sx - q = 0$. If the equation of the circle described on PQ as diameter is $2(x^2 + y^2) - 11x - 14y - 22 = 0$, then $2r + s - 2q + p$ is equal to _____.

Ans. **[7.00]**

Sol. Let $P(x_1, y_1)$ & $Q(x_2, y_2)$

$\Rightarrow 2x^2 - rx + p = 0 \begin{cases} x_1 \\ x_2 \end{cases}$

& $x^2 - sx + q = 0 \begin{cases} y_1 \\ y_2 \end{cases}$

$\therefore$ Equation of circle $\equiv (x - x_1)(x - x_2) + (y - y_1)(y - y_2) = 0$

$\Rightarrow x^2 - (x_1 + x_2)x + x_1x_2 + y^2 - (y_1 + y_2)y + y_1y_2 = 0$

$\Rightarrow x^2 - \frac{r}{2}x + \frac{p}{2} + y^2 + sy - q = 0$

$\Rightarrow 2x^2 + 2y^2 - rx + 2sy + p - 2q = 0$

Compare with $2x^2 + 2y^2 - 11x - 14y - 22 = 0$

We get $r = 11$, $s = 7$, $p - 2q = -22$

$\Rightarrow 2r + s + p - 2q = 22 + 7 - 22 = 7$

Q.85 The number of values of x in the interval $\left(\frac{\pi}{4}, \frac{7\pi}{4}\right)$ for which $14\text{cosec}^2x - 2\sin^2x = 21 - 4\cos^2x$ holds, is _____.

Ans. **[4.00]**

Sol. $\frac{14}{\sin^2 x} - 2\sin^2 x = 21 - 4(1 - \sin^2 x)$

Let $\sin^2 x = t$

$\Rightarrow 14 - 2t^2 = 21t - 4t + 4t^2$

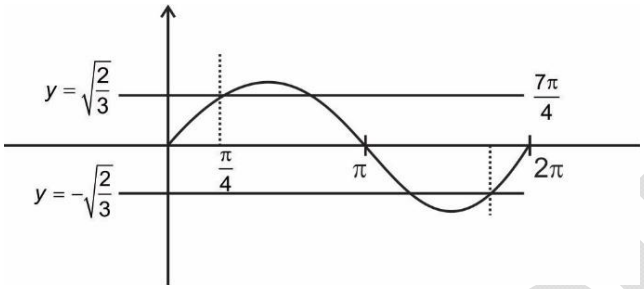
$\Rightarrow 6t^2 + 17t - 14 = 0$

$\Rightarrow 6t^2 + 21t - 4t - 14 = 0$

$\Rightarrow 3t(2t + 7) - 2(2t + 7) = 0$

$\Rightarrow \sin^2 x = \frac{2}{3}$ or $-\frac{7}{3}$ (rejected)

$\Rightarrow \sin x = \pm\sqrt{\frac{2}{3}}$



$\therefore \sin x = \pm\sqrt{\frac{2}{3}}$ has 4 solutions in $\left(\frac{\pi}{4}, \frac{7\pi}{4}\right)$

Q.86 For a natural number n, let $\alpha_n = 19^n - 12^n$. Then, the value of $\frac{31\,\alpha_9 - \alpha_{10}}{57\,\alpha_8}$ is _____

Ans. **[4.00]**

Sol. $\alpha_n = 19^n - 12^n$

Let equation of roots 12 & 19 i.e.

$x^2 - 31x + 228 = 0$

$\Rightarrow (31 - x) = \frac{228}{x}$ (where x can be 19 or 12)

$\therefore \frac{31\alpha_9 - \alpha_{10}}{57\alpha_8} = \frac{31(19^9 - 12^9) - (19^{10} - 12^{10})}{57(19^8 - 12^8)}$

$= \frac{19^9(31 - 19) - 12^9(31 - 12)}{57(19^8 - 12^8)}$

$= \frac{228(19^8 - 12^8)}{57(19^8 - 12^8)} = 4.$

Q.87 Let $f : R \to R$ be a function defined by $f(x) = \left(2\left(1 - \frac{x^{25}}{2}\right)(2 + x^{25})\right)^{\frac{1}{50}}$. If the function $g(x) = f(f(f(x))) + f(f(x))$, then the greatest integer less than or equal to g(1) is _____.

Ans. **[2.00]**

Sol. $f(x) = \left(2\left(\frac{2 - x^{25}}{2}\right)(2 + x^{25})\right)^{\frac{1}{50}}$

$= \left(4 - x^{50}\right)^{\frac{1}{50}}$

$$f(f(x)) = \left(4 - \left((4 - x^{50})^{\frac{1}{50}}\right)^{50}\right)^{\frac{1}{50}} = x$$

As f (f(x)) = x we have
$g(x) = f(f(f(x))) + f(f(x)) = f(x) + x$
$\Rightarrow g(x) = (4 - x^{50})^{1/50} + x$
$\Rightarrow g(1) = 3^{1/50} + 1$
$\Rightarrow [g(1)] = 2$

Q.88 Let the lines

$L_1 : \vec{r} = \lambda(\hat{i} + 2\hat{j} + 3\hat{k}), \lambda \in R$

$L_2 : \vec{r} = (\hat{i} + 3\hat{j} + \hat{k}) + \mu(\hat{i} + \hat{j} + 5\hat{k})$; $\mu \in R$,

intersect at the point S. If a plane $ax + by - z + d = 0$ passes through S and is parallel to both the lines L_1 and L_2, then the value of a + b + d is equal to_____.

Ans. **[5.00]**

Sol. As plane is parallel to both the lines we have d.r's of normal to the plane as < 7, –2, – 1 >

$$\left(\text{from} \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 2 & 3 \\ 1 & 2 & 5 \end{vmatrix} = 7\hat{i} - \hat{j}(2) + \hat{k}(-1)\right)$$

Also point of intersection of lines is $2\hat{i} + 4\hat{j} + 6\hat{k}$

$\therefore$ Equation of plane is
$7(x - 2) - 2(y - 4) - 1(z - 6) = 0$
$\Rightarrow 7x - 2y - z = 0$
$a + b + d = 7 - 2 + 0 = 5$

Q.89 Let A be a 3 × 3 matrix having entries from the set {–1, 0, 1}. The number of all such matrices A having sum of all the entries equal to 5, is _____.

Ans. **[414.00]**

Sol. Let matrix $\begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix}$

We need
$a + b + c + d + e + f + g + h + i = 5$

Possible cases	Number of ways
$5 \to 1$'s, $4 \to$ zeroes	$\frac{9!}{5!4!} = 126$
$6 \to 1$'s, $2 \to$ zeroes, $1 \to -1$	$\frac{9!}{6!2!} = 252$
$7 \to 1$'s, $2 \to -1$'s	$\frac{9!}{7!2!} = 36$

Total ways = 126 + 252 + 36 = 414

Q.90 The greatest integer less than or equal to the sum of first 100 terms of the sequence $\frac{1}{3},\frac{5}{9},\frac{19}{27},\frac{65}{81},.....$ is equal to _____

Ans. **[98.00]**

Sol. $S = \frac{1}{3} + \frac{5}{9} + \frac{19}{27} + \frac{65}{81} + ...$

$$= \sum_{r=1}^{100}\left(\frac{3^r - 2^r}{3^r}\right)$$

$$= 100 - \frac{2}{3}\,\frac{\left(1-\left(\frac{2}{3}\right)^{100}\right)}{1/3}$$

$$= 98 + 2\left(\frac{2}{3}\right)^{100}$$

$\therefore [S] = 98$

CAREER POINT